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RESEARCH ARTICLE

Transit Accessibility and Housing Prices: A Hedonic Model for an Urban Agglomeration

Evidence from the Opening of a New Light Rail Line

Artyom Melnikov[1]✉, **Pavel Gromov**[2], **Daria Polyakova**[1], **Evgeny Sedov**[2]

[1] Polis Graduate School of Urban Studies, Russia

[2] Meridian Institute for Economics and Spatial Development, Russia

✉ Corresponding author: Artyom Melnikov, a.melnikov@example.org

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ABSTRACT

Investment in public transport reshapes the spatial structure of a city, and one of the most visible manifestations of this is the change in housing prices near new stations. This article estimates the effect of opening a 17-km light rail line on apartment prices in an anonymised agglomeration with a population of about 1.2 million. The data comprise 46,000 transactions and sale listings for 2018–2024, geocoded and supplemented with building and neighbourhood characteristics. Transit accessibility was measured as the number of jobs reachable within 45 minutes by public transport before and after the opening of the line. The effect was identified with a hedonic difference-in-differences model in which the treatment group consists of apartments within 800 m of the new stations. After the line opened, apartment prices in the catchment area increased by 6.8% more than in the control area, and the effect began to appear as early as the announcement of construction. The effect is stronger for apartments in peripheral districts with initially low accessibility and practically absent in the centre. An increase in accessibility of 10,000 jobs is associated with a 1.9% increase in the price per square metre. The results can be used to justify land value capture mechanisms for financing transport infrastructure.

Keywords: transit accessibility; hedonic model; housing market; light rail; land value capture

Транспортная доступность и цены на жильё: гедонистическая модель для городской агломерации. На примере открытия новой линии скоростного трамвая

Артём Мельников[1]✉, *Павел Громов*[2], *Дарья Полякова*[1], *Евгений Седов*[2]

[1] Высшая школа урбанистики «Полис», Россия

[2] Институт экономики и пространственного развития «Меридиан», Россия

АННОТАЦИЯ

Инвестиции в общественный транспорт меняют пространственную структуру города, и одним из наиболее наглядных проявлений этого является изменение цен на жильё вблизи новых станций. В статье оценивается влияние открытия линии скоростного трамвая протяжённостью 17 км на цены квартир в условной агломерации с населением около 1,2 млн человек. Использованы данные о 46 тыс. сделок и объявлений о продаже за 2018–2024 гг., геокодированные и дополненные характеристиками домов и окружающей среды. Показатель транспортной доступности рассчитан как число рабочих мест, достижимых за 45 минут на общественном транспорте, до и после открытия линии. Для идентификации эффекта применена гедонистическая модель с методом разности разностей, в которой экспериментальной группой служат квартиры в радиусе 800 м от новых станций. Установлено, что после открытия линии цены квартир в зоне влияния выросли на 6,8 % больше, чем в контрольной зоне, причём эффект начал проявляться уже после объявления о строительстве. Эффект сильнее для квартир в периферийных районах с исходно низкой доступностью и практически отсутствует в центре. Рост доступности на 10 тыс. рабочих мест связан с увеличением цены квадратного метра на 1,9 %. Результаты могут использоваться для обоснования механизмов возврата части прироста стоимости земли на финансирование транспортной инфраструктуры.

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Ключевые слова: транспортная доступность; гедонистическая модель; рынок жилья; скоростной трамвай; возврат стоимости земли

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1 INTRODUCTION

The relationship between transport accessibility and property values is one of the fundamental themes of urban economics. The classic monocentric city model predicts that a reduction in transport costs raises the value of plots located far from the centre. Empirical studies conducted in various countries generally confirm this, but estimates of the effect vary considerably.

The opening of a new light rail line in the agglomeration under study in 2022 created a natural experiment. The aim of the study is to estimate the resulting change in housing prices, trace its dynamics over time and space, and relate it to the change in job accessibility.

2 MATERIALS AND METHODS

The source data comprise 18 thousand registered transactions and 28 thousand listings of flats for sale for 2018–2024. For listings, the last asking prices before removal from publication were used. All properties were geocoded and supplemented with characteristics: floor area, floor, year of construction and building material, and distance to the centre, parks and schools.

Job accessibility was calculated from public transport timetables and data on the location of employment. The hedonic model for the logarithm of the price per square metre includes property characteristics, district and quarter-period fixed effects, and the interaction of the impact zone indicator (up to 800 m from a station) with the “announcement”, “construction” and “operation” periods.

3 RESULTS

The dynamics of the price difference between the impact zone and the control zone are shown in Fig. 1. Before the construction announcement, the differences are statistically insignificant, which supports the parallel trends assumption. After the announcement in 2019, the difference begins to grow and reaches 6.8% in the operation period.

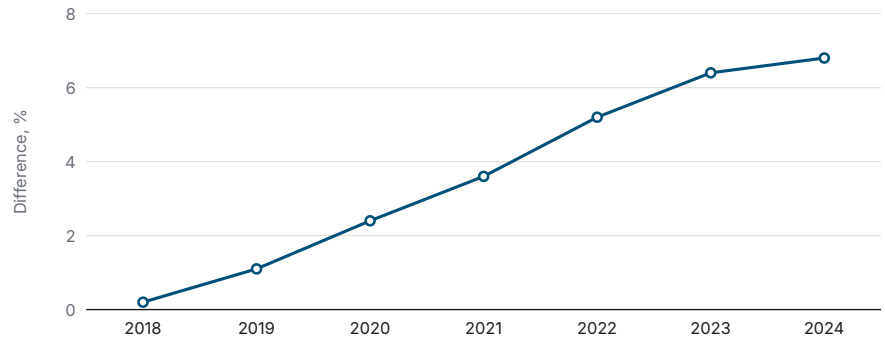


Fig. 1. Difference in price per square metre between the impact zone and the control zone by year, %

Estimates of the main model and its modifications are given in Table 1. The effect for peripheral districts is almost three times larger than for the middle zone, while in the centre it does not differ statistically from zero.

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Table 1. Estimates of the hedonic difference-in-differences model (dependent variable: log price per m²)

Specification	Effect, %	Std. error	N
All districts	6,8	1,3	46 112
Periphery	11,2	2,1	17 840
Middle zone	4,1	1,6	19 305
Centre	0,9	2,4	8 967
Transactions only	6,1	1,8	18 024

4 DISCUSSION

The magnitude of the effect lies in the middle of the range typical of international studies of rapid rail transit. The rise in prices even before the line opened indicates that the market capitalises expectations, which is important for the timing of land value capture mechanisms.

The heterogeneity of the effect across zones is consistent with theory: where accessibility was already high, the new line adds little. A limitation of the study is the use of asking prices for part of the sample; however, the estimate based on transactions only is close to the main one.

5 CONCLUSION

The opening of the light rail line raised housing prices in the impact zone by 6.8%, with the largest effect observed on the periphery. This provides a rationale for using instruments that capture part of the value uplift to finance transport projects.

REFERENCES

1. Rosen S. Hedonic prices and implicit markets: product differentiation in pure competition // Journal of Political Economy. 1974. Vol. 82, no. 1. P. 34–55.
2. Hansen W. G. How accessibility shapes land use // Journal of the American Institute of Planners. 1959. Vol. 25, no. 2. P. 73–76.
3. Мельников А. С. Метрики транспортной доступности на основе расписаний общественного транспорта // Геоинформатика и общество. 2023. № 2. С. 5–21.
4. Громов П. А., Седов Е. В. Капитализация инфраструктурных инвестиций в цены жилья: обзор эмпирических исследований // Пространственная экономика и политика. 2022. № 4. С. 9–30.
5. Debrezion G., Pels E., Rietveld P. The impact of railway stations on residential and commercial property value: a meta-analysis // Journal of Real Estate Finance and Economics. 2007. Vol. 35, no. 2. P. 161–180.
6. Полякова Д. И. Анонсирование инфраструктурных проектов и ожидания рынка жилья // Городские исследования. 2024. Т. 8, № 1. С. 44–61.
7. Ng W., Carvalho F. Land value capture for light rail: international practice // Transport Finance Review. 2021. Vol. 10. P. 71–90.
8. Мельников А. С., Полякова Д. И. Геокодирование объявлений о продаже жилья: методы и ошибки // Прикладная аналитика. 2023. Т. 7, № 4. С. 63–79.

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